1 Saving the activation changes the compiled program, so the numbers come out different



No probe — normalize + rotate (one GPU kernel)

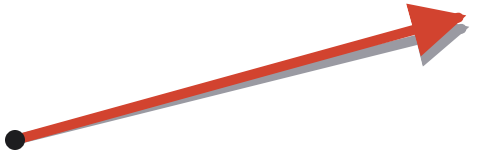


Probe — normalize — rounded to BF16 — rotate

Earlier layers also lay out memory differently.

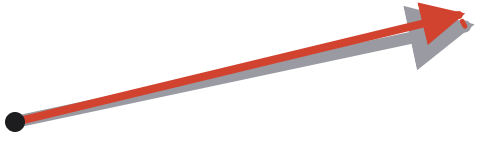
2 The user token's hidden state ends up 2.5% different, the same way in every prompt

`<|im_start|>` `user` Is a read-only probe … ?



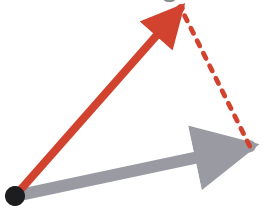
3 Head 6's key for user is short, so the same change turns it; other keys barely move

a typical head
key length 64.5
probe changes it by 1.8



turns 1.5°

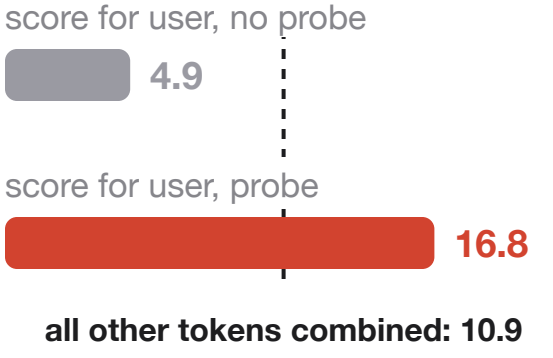
head 6 (trained)
key length 6.6
probe changes it by 4.2
drawn 5× larger



turns 36.6°

Qwen3 rescales every key to the same length before scoring, so only a key's direction counts.

4 The turned key matches head 6's query: user outscores every other token, the answer is Yes



head 6 reads

`</think>` → **No**

`user` → **Yes**